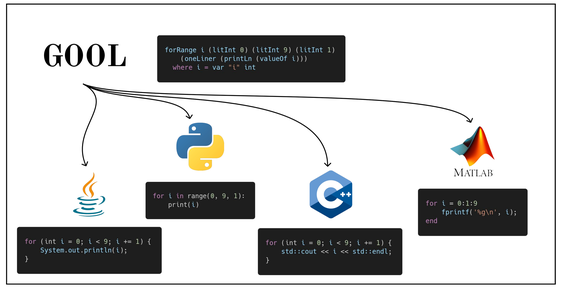
A Generic Representation of Program Organization

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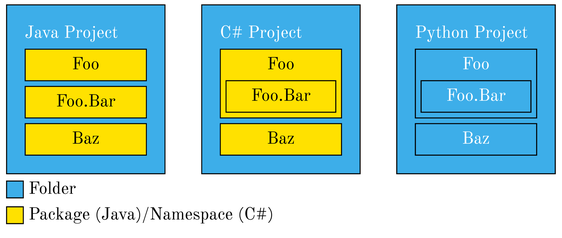
What Is GOOL? [1]

- **G**eneric **O**bject-**O**riented **L**anguage
- "Write once, generate everywhere"



Program Organization

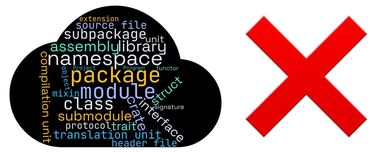
- Different languages have different systems for organizing program logic
- These systems also have different visibility rules



Problem: current GOOL doesn't know about these differences!

- Only knows how to generate code fitting the lowest common denominator of these systems
- This can cause it to generate ugly or broken code for some languages

Generically Representing Program Organization in GOOL



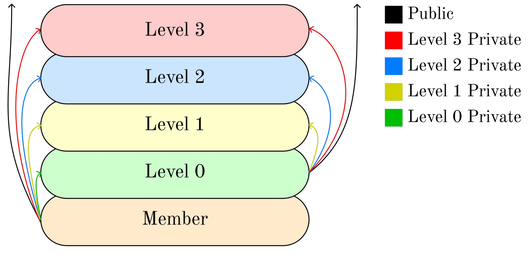
Indexed Abstraction Levels

"Package", "module", "namespace" etc. are over-used, so instead we use indexed levels of abstraction.

GOOL	Java	Swift	MATLAB
Level 3	Project	Module	Project
Level 2	Package	Folder	Namespace
Level 1	File with a static class	File	Folder
Level 0	Static Method	Function	File with a Function

Visibility

- Organizational levels form a tree
- Allow each level to be visible within the closest enclosing level
- Many visibilities will not be expressible in most languages
  - Choose the closest visibility that preserves semantics
  - If no visibilities preserve semantics, throw an error for that language



Remaining Questions

- Abstract vs. concrete program structures
- How to guarantee preservation of semantics and intent?
- Control of organization: GOOL vs. the programmer

References

- [1] Carette, J., MacLachlan, B., Smith, S. (2020). GOOL: A Generic Object-Oriented Language. PEPM '20. https://dl.acm.org/doi/10.1145/3372884.3373159

Acknowledgements

- Mention OGS and NSERC?